

MB 301: Problem Solving and Decision Making for Managers

Course Code: MB 301

Course Name: Problem Solving and Decision Making for Managers

L-T-P-C: 3-0-0-3

Prerequisites: NA

Intended for: IMBA

Mutual Exclusion: NA

Approval: 59th BoA (I-MBA curriculum); course proposal approved in the 60th BoA

Preamble

Normally, thinking skills are based on some structure such as critical thinking, constructive thinking, lateral thinking, creative thinking, vertical thinking, parallel thinking, etc. However, all these approaches concentrate more on “talking about problems” rather than “solving problems.” Thus, it becomes necessary for the students to connect thinking and problem-solving skills to tackle complex, nonlinear, and uncertain problems in the real world. This course provides what, why and how of thinking skills through various toolsets, proper mindsets, and appropriate skillsets. This course includes discussions and brainstorming through a range of decision-making tools & techniques, brain teasers, games, and puzzles. Further, it consists of discussions on heuristics for tackling these types of problems, along with visualization of the contrasting nature of problem-solving in humans and computers.

Course Modules

- **Introduction:** Problems and goals; Understanding problem solving and decision making, Types of problems and decision making, rules of analytical problem solving, Alternatives, Objectives, Criteria. (2)
- **Mind:** The mind and the brain, the agents of the mind, consciousness, and memory; Cognitive processes: perception, attention, Intuition, emotion, thinking, intelligence, knowledge representation, categories, learning, motivation, judgment, reasoning, creativity, beliefs, biases and errors; Critical and creative thinking, Design Thinking, Common sense and mental models. (4)
- **Creativity and Creative People:** Creativity contexts, creativity realms and limitations, Creativity and Uncertainty, creative process and people; motivation and creativity, Creativity Techniques- convergent and divergent thinking; idea generation; value proposition canvas, SCAMPER.; Creativity for Decision Making - Brainstorming, Synectics, Morphological Analysis, and Lateral thinking; Creativity and inspiration, creativity and evolution, creativity in a nonlinear world, inventions, and innovators. (5)
- **Problem Identification and Definition:** Critically thinking on problems, Problem identification: Process, methods, and practices; difficulties and errors; improvement; Problem Definition: Defining managerial problems, Techniques- Affinity diagram, Concept Maps, Stratification, Pareto Analysis; Definitional Mistakes- Framing Errors, Focusing Errors, Missed Connections, The Wrong elements and relationships; Situation Definition and Analysis Method (SDA). (5)
- **Problem Diagnosis:** Causality and its types; Diagnosis- Importance and Strategies; Diagnostic Process- Data and information gathering, hypothesis generation and testing, diagnostic errors; Diagnostic Methods: Diagnostic and Solution Maps, Experimentation, Cause-and-Effect Diagram, Concentration Diagram, Why-Why Diagram, Kepner-Tregoe Method, Root Cause Analysis- Determine the Actual Causes. (5)

- **Problem Solutions:** Models and Modelling; Solution Space- feasible and infeasible; Identify Potential Solutions- Solution Definition and Solution Maps, Alternative Generation, Alternative Generation Heuristics, brainstorming for alternative generation and removing unsuitable alternatives; Analysing Critical Assumptions, The Problem Analytic Strategy, Analytic Bootstrapping Creativity (ABC) Method. (5)
- **Solution Evaluation and Implementation:** Performance metrics and Key Performance Indicators (KPIs), Brainstorming for Criteria and Alternatives of choice, Morphological Analysis for arranging ideas and the connections in a decision hierarchy, Evaluating Alternatives, structuring decisions, Judgements and comparisons, decision tree, Paired Comparison Analysis, Analytic hierarchy and network process, Group evaluation, Six Thinking Hats method, Solution Implementation and monitoring. (4)
- **Estimation, Simulation and Optimization:** Estimation and its role in Problem-solving and Decision-Making; Simulation and its types, the importance of simulation and optimization for problem-solving and decision-making, the role of data and models in simulations. physical simulation vs. computer simulation. (4)
- **Analytical Thinking and Creativity:** Analytical Thinking and decision making, stages of analytical thinking, data analytic thinking, Creative Analytical Thinking; Pattern finding and Analytical creativity; Capability development of Quantitative Analysts, Quantitative Analysts and Business Decision Makers. (4)

Laboratory/Practical/Tutorial Modules

NA

Text Books

Not Prescribed

References

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7. G. Polya (1988), *How to Solve It*, Princeton University Press.
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9. Jeanne Liedtka, Andrew King and Kevin Bennett (2013), *Solving Problems with Design Thinking*, Columbia University Press.

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 13. J. Butterfield (2010), *Problem Solving and Decision Making*, Cengage Learning.
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